Updating the Estimation of Consumption Elasticities

A note on consumption elasticities in MAKRO

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June 2026

1 Updating the consumption elasticity estimates

This note updates the estimates of the consumption elasticity. The purpose of the exercise is to assess whether the existing substitution elasticity estimates from Kronborg and Kastrup (2021), can be improved by estimating the relative demand equation jointly with the CES aggregation equation.

In Kronborg and Kastrup (2021), the elasticities are estimated for a nested CES consumption system using a relative demand equation and an unobserved preference component estimated with a Kalman filter.

The new approach is inspired by the system-estimation idea in Kastrup and Vikkelsø (2023). The key idea is to estimate the relative first-order condition jointly with the CES aggregation equation for each consumption nest.

The analysis is carried out in three steps. First, the MAKRO consumption data are organised into the CES nests used by Kronborg and Kastrup (2021). Second, the relative demand equation and the CES aggregation equation are estimated jointly in a baseline system. Third, the preference specification is extended through a structured non-Kalman model and a direct Kalman system.

2 Data

The estimation is based on consumption data extracted from the MAKRO database. For each consumption component, the dataset contains annual quantities, prices and expenditure values. The sample is initially restricted to 1983-2017 in order to compare the new estimates with the previous estimates.

The data are organised according to the nested CES structure used in the Kronborg and Kastrup (2021). For each nest, aggregate price and quantity series are constructed using Paasche price indices. The resulting dataset contains the two input components in each nest, the observed aggregate consumption series and the calibrated CES share parameter.

3 Estimation strategy

For each CES nest, the objective is to estimate the substitution elasticity, denoted by σ. The two consumption inputs are denoted by $C_{1,t}$ and $C_{2,t}$, with corresponding prices $P_{1,t}$ and $P_{2,t}$. The aggregate consumption level in the nest is denoted by C_t, while ω is the CES share parameter and μ_t captures time-varying relative preferences.

The starting point is the relative demand equation used in Kronborg and Kastrup (2021):

$$\ln\left(\frac{C_{1,t}}{C_{2,t}}\right) = \alpha - \sigma \ln\left(\frac{P_{1,t}}{P_{2,t}}\right) + (\sigma - 1)\mu_t + \varepsilon_{1,t}. \tag{1}$$

This equation identifies the substitution elasticity from relative prices and relative quantities. The term μ_t captures shifts in relative preferences between the two goods. In Kronborg and Kastrup (2021), μ_t was estimated using a Kalman filter.

The same first-order condition is often written in an equivalent budget-share form:

$$\log\left(\frac{P_{1,t}C_{1,t}}{P_{2,t}C_{2,t}}\right) = \log\left(\frac{\omega}{1-\omega}\right) + \frac{\sigma-1}{\sigma}\left[\mu_t + \log\left(\frac{C_{1,t}}{C_{2,t}}\right)\right] + \varepsilon_{1,t}. \tag{2}$$

The parameter ω is related to the constant term in the relative demand equation.

The main update is to add the CES aggregation equation to the estimation system, which intention is to improve the identification of σ:

$$C_t = \left[\omega\left(e^{\mu_t}C_{1,t}\right)^{\frac{\sigma-1}{\sigma}} + (1-\omega)C_{2,t}^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}. \tag{3}$$

This implies that the same preference component μ_t must explain both relative demand and aggregate consumption.

Several model variants are estimated. The baseline model, estimates the relative demand equation and the CES aggregation equation jointly with fixed calibrated ω. In the baseline model, preferences are represented by a simple linear trend.

$$\mu_t = a + bt \tag{4}$$

In the structured and direct Kalman specifications, the preference component is allowed to follow a more flexible process where it is decomposed into a trend component, τ_t, and a persistent transitory component, c_t:

$$\begin{aligned} \mu_t &= \tau_t + c_t, \\ \tau_t &= \tau_{t-1} + \eta_t, \\ c_t &= \rho c_{t-1} + \nu_t. \end{aligned} \tag{5}$$

The trend component follows a random walk, while the transitory component follows an autoregressive process. The parameter ρ_c measures the persistence of the transitory component. The shocks η_t and ν_t allow the trend and transitory components to change over time.

The structured model estimates the preference path directly by minimizing the joint model errors, while the direct Kalman system estimates the same process using a Kalman filter and maximum likelihood.

4 Comparison of estimated elasticities

Table 4.1 compares the substitution elasticities obtained from the different estimation approaches. The baseline and structured non-Kalman estimates are broadly in line with the estimates reported by Kronborg and Kastrup (2021) for most nests.

The Kalman variants are less stable. Several estimates are close to the imposed parameter bounds, while the remaining estimates often differ noticeably from those reported by Kronborg and Kastrup (2021). This indicates that the full Kalman specification is more sensitive to the exact model setup than the baseline and structured non-Kalman specifications.

Table 4.1
CES substitution elasticity estimates, 1983–2017

Nest	Kronborg and Kastrup (2021)	Baseline	Structured	Kalman
Tourism vs. services	1.250	1.101	1.102	0.603
Tourism/services vs. goods	0.940	1.145	1.153	1.140
Tourism/services/goods vs. energy	0.260	0.285	0.283	2.500
Non-housing excl. cars vs. cars	1.040	1.052	1.052	1.052

Note: "Baseline" estimates the relative demand equation jointly with the CES aggregation equation, using fixed calibrated omega and a simple linear preference trend. "Structured" adds a structured preference process without Kalman filtering. "Kalman" uses the direct Kalman system with fixed omega and equal measurement variance across the two equations. Boundary estimates at 0.100 or 2.500 indicate weak identification in that model version.

5 Extended sample

The estimation sample is extended to include the latest available MAKRO data. Table 5.1 reports the estimated substitution elasticities for the three model specifications using the extended sample.

The extended-sample results suggest that the estimates are robust to including the latest available MAKRO data. The three model specifications give very similar substitution elasticities across all nests. This is especially clear for the tourism/services, goods and car nests. The energy nest also remains close to the estimate obtained on the shorter sample.

Compared to the 1983–2017 results, the direct Kalman system is considerably more stable in the extended sample and no longer produces boundary estimates.

Table 5.1
CES substitution elasticity estimates, 1983–2025

Nest	Baseline	Structured	Kalman
Tourism vs. services	1.101	1.104	1.102
Tourism/services vs. goods	1.159	1.187	1.156
Tourism/services/goods vs. energy	0.283	0.280	0.281
Non-housing excl. cars vs. cars	1.053	1.054	1.053

Note: "Baseline" estimates the relative demand equation jointly with the CES aggregation equation, using fixed calibrated ω and a simple linear preference trend. "Structured" adds a structured preference process without Kalman filtering. "Kalman" uses the direct Kalman system with fixed ω and equal measurement variance across the two equations.

6 Conclusion

This note updates the estimation of the consumption elasticities by estimating the relative first-order condition jointly with the CES aggregation equation. Compared to the previous estimation method, this imposes an additional restriction on the preference component, since preferences must be consistent with both relative demand and aggregate consumption.

The baseline and structured specifications produce estimates that are broadly in line with the previous DREAM estimates for the 1983–2017 sample. This suggests that the joint system approach can reproduce the main results from the earlier consumption paper while adding more structure to the estimation.

The direct Kalman system gives less stable results on the 1983–2017 sample. Although some nests produce plausible estimates, the energy nest reaches the upper parameter bound, indicating weak identification in the full state-space specification. However, when the sample is extended to 2025, the Kalman estimates become much more stable and are closely aligned with the baseline and structured specifications.

Taken together, the results support using the joint system approach as a basis for updating the consumption elasticities. However, further testing is needed before the full Kalman specification can be used as the preferred model.

7 Appendix

Tables 7.1–7.3 report the main estimation output for the three new model specifications: the baseline system, the structured preference model, and the direct Kalman system.

Table 7.1
Baseline estimation results, 1983–2017

	cTurTje	cTurTjeVar	cTurTjeEne	cIkkeBol
σ	1.10	1.14	0.285	1.05
ω	0.153	0.516	0.998	0.941
a	-2.76	0.882	0.199	0.199
b	0.0259	0.00520	-0.0134	0.000336
SSR	0.363	0.705	1.45	1.71
Converged	TRUE	TRUE	TRUE	TRUE

Note: The baseline model estimates the relative demand equation and the CES aggregation equation jointly. The CES share parameter ω is fixed at its calibrated Paasche value. Preferences are represented by a linear trend, $\mu_t = a + bt$. SSR denotes the sum of squared residuals from the joint estimation.

Table 7.2
Structured preference estimation results, 1983–2017

	cTurTje	cTurTjeVar	cTurTjeEne	cIkkeBol
σ	1.10	1.15	0.283	1.05
ω	0.153	0.516	0.998	0.941
ρ_c	0.902	0.915	-0.388	0.565
SSR	0.279	0.703	0.669	1.71
Converged	TRUE	TRUE	TRUE	TRUE

Note: The structured model estimates the relative demand equation and the CES aggregation equation jointly, while allowing the preference component to follow a more flexible structured process. The CES share parameter ω is fixed at its calibrated Paasche value. SSR denotes the sum of squared residuals from the joint estimation.

Table 7.3
Direct Kalman system estimation results, 1983–2017

	cTurTje	cTurTjeVar	cTurTjeEne	cIkkeBol
σ	0.603	1.14	2.50	1.05
ω	0.153	0.516	0.998	0.941
ρ_c	0.980	0.980	0.0322	0.0325
$\mathrm{var}(\eta)$	0.00000239	0.000000325	0.00000182	0.000000197
$\mathrm{var}(\nu)$	0.000000391	0.000000117	0.000000267	0.000000375
$\mathrm{var}(\varepsilon)$	0.284	0.0104	9.20	0.0252
Negative log likelihood	61.0	-52.0	181	-20.5
Converged	TRUE	TRUE	TRUE	TRUE

Note: The direct Kalman system estimates the time-varying preference component in a state-space model. The relative demand equation and the CES aggregation equation are included directly in the measurement system. The CES share parameter ω is fixed at its calibrated Paasche value, and the two measurement equations share a common measurement variance, $\mathrm{var}(\varepsilon)$. The reported likelihood value is the negative log likelihood.

References

Kastrup, C. B., & Vikkelsø, C. (2023). Overcoming bias in elasticity of substitution estimation: A novel approach with empirical application. *DREAM - Danish Reasearch Institute for Economic Analysis and Modelling*.

Kronborg, A. F., & Kastrup, C. S. (2021). Estimering af forbrugssystemet i makro. *DREAM - Danish Reasearch Institute for Economic Analysis and Modelling*.